

Python for Data Science - Notes

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Introduction

Interpreter vs. Compiler Languages

An interpreter translates code and sends it to the CPU. The code is sent to the interpreter line by line. The interpreter provides a console for ad-hoc commands. It's usually slower than a compiler and not good for optimization.

A compiler processes source code and returns an executable file. Compiler languages are usually fast and made for optimization (speed, storage use). Executable code is almost impossible to reverse-engineer.

Intelligent Development Environment (IDE)

Typically, an IDE provides additional tools to make programming more convenient. IDEs provide tools such as:

- Debuggers
- Deployment chain control
- File browser
- Terminal
- Version control interface
- etc.

Note: Python is a text-only language that can be written in a simple text editor.

Python console vs. script

The Python console is a direct interface to the interpreter. Every instruction is sent sequentially to the interpreter. Commands sent in the console are sent directly to the interpreter.

A Python script is a collection of commands. Executing a Python script means sending all the commands it contains consecutively to the interpreter.

Virtual environments (venv)

To avoid dependency issues of packages and keep the development environment clean, developers programme applications in virtual environments.

```
Python3 -m venv .venv # Standard name
```

Note: It is best practice to name the virtual environment `.venv` to hide it in the file system.

You need to activate the virtual environment.

```
. .venv/bin/activate # For Linux
```

You can install packages into the venv with `python3-pip`.

Python Basics

Variabels and Basic data types

A variable is a name that refers to a particular or undefined value. In programming languages, we use them as a reference to a particular storage location. A variable always consists of a name, a data type, a storage location and a value. Python does not require any type for variable definitions. It assumes the type from the value.

Note: Variable names are usually written in lowercase.

Basic data types:

- Integer (int)
- String (str)
- Boolean (bool).
- List
- Dictionary

Mathematical Operations

Depending on the context, two particular mathematical operators can have different meanings. The sum operator (+), except for numerical values, means appending one element to another. This usually requires two elements of the same data type. The multiplication operation (*) always has to be applied with an integer.

Note: Sum and multiplication can be used with strings.

Boolean operations

Two Boolean values can be combined in different ways using the keywords **and**, **or** and **xor**. Any Boolean operation can be inverted using the keyword **not**.

Sequential data types

In Python, sequential data types can comprise mixed data types and can have multiple dimensions.

- List: Ordered, changeable.
- Tuple: Ordered, unchangeable.
- Dictionary: Key-Value-Pairs.
- Set: No Duplicates.

Type casting

Depending on the operation, the same data may have to appear in different types. Data of a particular type can be transformed into a different data type.

```
x = 3

# Same variable but different data type
print("x as int= ", int(x))
print("x as float= ", float(x))
print("x as string= ", str(x)) # Looks like int but is type str!
```

```
x as int= 3
x as float= 3.0
x as string= 3
```

Input and Output

Interaction with an application requires input and output. Use the **input()** function to print the passed string to the standard output (i.e., terminal). The input is read from the terminal and converted to a string.

Boolean

A boolean expression can be of two states only (**True** or **False**).

Note: Python treats all that is NOT 'empty', '0', 'False' or 'None' as 'True'

Priority-List:

1. not
2. and
3. or

Bitwise operations

Bitwise operations in Python are used to manipulate individual bits of integer values.

- `x | y` bitwise or of x and y
- `x ^ y` bitwise exclusive or of x and y
- `x & y` bitwise and of x and y
- `x << n` x shifted left by n bits
- `x >> n` x shifted right by n bits
- `~x` the bits of x inverted

Comparison operations

- `<` strictly less than
- `<=` less than or equal
- `>` strictly greater than
- `>=` greater than or equal
- `==` equal
- `!=` not equal
- `is` object identity
- `is not` negated object identity

Note: Comparisons can be chained arbitrarily.

Indentation and syntax

Block operations start with a colon (:) and are defined by indentations. Indentations can have an arbitrary number of spaces but must be constant for all instructions within the same block.

Single block

```
instruction
instruction
block header:
....block instruction
....block instruction
....block instruction
instruction
instruction
```

Nested blocks

```
instruction
instruction
block 1 header:
....block 1 instruction
....block 2 header:
.....block 2 instruction
.....block 2 instruction
....block 1 instruction
....block 1 instruction
instruction
```

If-else condition

The if-else statement is a control structure for checking a condition, allowing you to execute different code blocks depending on whether the condition is met.

```
x = 10

if x == 10:
    print("x has the value 10")
else:
    print("x does not have the value 10")
```

x has the value 10

Elif concatenation

elif allows for concatenating multiple conditions.

```
x = 6

if x == 10:
    print("x has the value 10")
elif x % 2 == 0:
    print("x is an even number")
```

x is an even number

Shorthand statement

Shorthand if statements allow for less code and better readability.

```
x = 10

# Block
if x == 10:
    x += 1
    print(x)
else:
    None

## Is the same as

x = 10
# One Line
x = x + 1 if x == 10 else None
print(x)
```

11
11

Match case

Instead of using multiple combined conditions with elif statements, an expression can be directly checked against multiple specific cases.

```
x = 2
match x:
    case 0:
        print("number is 0")
    case 1:
        print("number is 1")
    case 2:
        print("number is 2")
    case _: # Default case
        print("number unknown")
```

number is 2

Loops

range ()

The class range() allows creating sequence objects with constant step sizes. Ranges implement all of the common sequence operations except concatenation and repetition.

Notes: The stop element is not included.

```
my_range = list(range(2, 11, 2)) # Start = 2, End = 11, Steps = 2
print(my_range)
```

[2, 4, 6, 8, 10]

len()

len() returns the number of elements in a sequence. Returns positive integer: 0 indicates an empty sequence.

```
my_lenght = len(range(1, 11, 1))
print(my_lenght)
```

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Sequence Slicing

Extracting sub-sequences of larger data containers is an important and often used operation. Slicing options are:

- By integer (particular element)
- By range or slice object

```
my_list = list(range(1, 11, 1))
print(f"My list: {my_list}")

# Extract only even numbers
even_numbers = my_list[1:11:2] # Start = Index 1 (second entry), Steps = 2
print(f"All even numbers in my list: {even_numbers}")

# Extract only odd numbers
odd_numbers = my_list[0:11:2] # Start = Index 0 (first entry), Steps = 2
print(f"All odd numbers in my list: {odd_numbers}")
```

```
My list: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
All even numbers in my list: [2, 4, 6, 8, 10]
All odd numbers in my list: [1, 3, 5, 7, 9]
```

Note: Same as `slice()`

for-loop

Loops are used to iterate over sequence objects by providing each element one after the other through a loop variable. Assign each element of the sequence one after another to the loop variable.

```
# Single loop
my_list = list(range(1, 11, 1))
print(f"My list: {my_list}")

for element in my_list:
    print(f"Element is {element}")
else:
    print("No more elements in my list")
```

```
My list: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Element is 1
Element is 2
Element is 3
Element is 4
Element is 5
Element is 6
Element is 7
Element is 8
Element is 9
Element is 10
No more elements in my list
```

Note: The loop variable is often called `i`: `for i in my_list:`.

while-loop

A while loop executes the loop body as long as the condition equals `True`. The while loop checks the condition each time before re-executing the loop body and terminates as soon as the condition is `False`.

```
aim = 5
counter = 0

while counter != aim: # != means not equal
    print(f"Counter at value {counter}")
    counter += 1 # Add 1 to the counter
else:
    print("Aim reached")
```

```
Counter at value 0
Counter at value 1
Counter at value 2
Counter at value 3
Counter at value 4
Aim reached
```

Note: Check whether the condition can be reached! Otherwise, you will end up in an infinite loop.

break, continue and pass

Python has three keywords to control the loop (and function) process flow:

break

Immediately break the current loop.

```
aim = 5
counter = 0

while counter != aim:
    print(f"Counter at value {counter}")
    counter += 1
    break # Force a loop break
else:
    print("Aim reached")
```

Counter at value 0

continue

Ignoring the rest of the loop body and jumping back to the header.

```
aim = 5
counter = 0

while counter != aim:
    print(f"Counter at value {counter}")
    counter += 1
    continue
    print("This msg will not be printed")
else:
    print("Aim reached")
```

Counter at value 0
Counter at value 1
Counter at value 2
Counter at value 3
Counter at value 4
Aim reached

pass

No operation. Regular iteration with no execution.

```
aim = 5
counter = 0

while counter != aim:
    print(f"Counter at value {counter}")
    counter += 1
    pass
else:
    print("Aim reached")
```

```
Counter at value 0
Counter at value 1
Counter at value 2
Counter at value 3
Counter at value 4
Aim reached
```

Enumerate

The enumerate returns an iterator that returns a tuple with an incrementing number for each element of the sequence.

```
x = list(range(0, 11, 2))

for idx, i in enumerate(x):
    print(f"Value {i} at Index {idx}")
```

```
Value 0 at Index 0
Value 2 at Index 1
Value 4 at Index 2
Value 6 at Index 3
Value 8 at Index 4
Value 10 at Index 5
```

SW05: Python Debugging

Remote App Development

Modern data analysis methods (e.g. deep learning, optimisation) require large computational resources such as memory or gpu. Remote resources can be provided by individual institutions like HSLU, or rent from web service providers like Amazon Web Services (AWS) or Google Cloud Platform (GCP). Data scientists use two popular Python development environments for data analysis and reporting: Desktop IDE (PyCharm) and Notebooks (Jupyter).

Jupyter Notebooks

A Jupyter Notebook is a web-based, open-source tool that combines live code (like Python), equations, output, and narrative text (Markdown) into a single, interactive document. It is organized into cells and is a cornerstone of data science for combining execution and documentation, making analysis transparent and reproducible.

Debugging

- Semantic errors: violating rules of coding language.
- Syntax errors: missing code elements (e.g. parenthesis).
- Logical errors: correct syntax but incorrect directions causing undesired output.
- Runtime errors: error happens when application is running or starting up.
- ect.

Breakpoints

Breakpoints define code locations where the execution shall stop. The execution is stopped before the selected line of code.

Debugger

A debugger is a tool that allows developers to meticulously examine and control the execution of their code. It enables the setting of breakpoints, allowing the developer to inspect the current state of variables, the call stack (the sequence of function calls that led to the current point), and memory. This step-by-step execution, often called stepping (e.g., step over, step into, step out), is fundamental for isolating and understanding the root cause of bugs or unexpected behavior.

Enum data type

An Enum is a set of symbolic names bound to unique values. Enumeration requires the package Enum.

```
from enum import Enum
```

Enum allows handle a set of values:

- days of the week
- Colors
- ect.

SW06: Functions, Strings & Files

Functions

Functions allow to combine multiple instruction into a function block that can be executed multiple times. Main advantages of functions are modularity (mitigates code duplication) and readability. Functions tackle only **one** particular issue at once. Ideally operate on its input only and produce some output.

Note: Functions are named by convention in snake case (i.e. lower case separated with '_'): `my_function()`.

The function header is made up of the function name and any optional parameters.

Note: Functions in Python are defined by the term **def**.

```
# Name = my_function
# Parameters: parameter1
def my_function(parameter1):
    pass # Do nothing, just pass.
```

The body of a function contains a sequence of operations and should always have an output value. The number of operations within a function is unlimited, and they can call other functions.

```
def my_function(parameter1):
    print(parameter1) # Call the print function
    return True

my_function("Test run") # Call my_function
```

Test run

True

Return Values

All functions (including purely functional) have one return value; at least **None**. Return value is a pointer to a storage location.

Documentation

Python offers a centralized documentation with docstrings; a built-in attribute assigned to each function named `__doc__`. If the `__doc__` attribute is set with a documentation string, it is callable by means of the `help()` function. You can place a string block (triple-quotes `"""doc text"""`) immediately next to the function definition to define the docstring.

```
def my_function(parameter1):  
    """ Prints a given parameter to the console """ # Docstring of my_function  
    print(parameter1) # Call the print function  
    return True
```

String Formatting

Python treats a string as a sequence (list) of characters (single letters, symbols or escape characters). Strings can be sliced, concatenated, allows index based access and formatting based on character position.

li :	'H'	'e'	'l'	'l'	'o'	' '	'w'	'o'	'r'	'l'	'd'	'\n'
idx:	0	1	2	3	4	5	6	7	8	9	10	11

Figure 1: String as a sequence of characters

Formatting strings means defining the representation and treatment of particular sequences.

f-Strings

An f-string can be used to add variables to a string. Rather than writing the string again every time the variable changes, we can use the f-string to add multiple variables to the string.

```
for i in range(1, 5):  
    # Using f-String  
    print(f"i is {i}") # i changes over time
```

```
i is 1  
i is 2  
i is 3  
i is 4
```

File Handling

Files enable storing data outside of the application and hence to keep the information for next execution or share data between different applications. Python accesses files through a file object.

```
file_object = open(file_name, mode)
```

! Important

The buffer results in unfinished writing process as long as the file stream (file_object) is not flushed by 'file_object.flush()' or file_object.close()'.

SW07: Functions part II Recursion

Namespace

Namespace is the definition of the visibility of a unique name for every single object (variables or methods). Object names must only be unique within a given namespace. Hence, the global namespace can have multiple local namespaces having objects with the same naming. The scope of an object refers to the code section from which an object is accessible.

- built-in: encompasses no programmer defined objects. Ends when the application ends.
- global: programmer defined objects available across the whole script. Ends when the module (i.e. script) is unloaded or the application ends.
- local: programmer defined objects in function blocks. Ends when the function (i.e. block) has been finished.

Local and Global Variables

Python allows creating variables without any restricted visibility. These variables in the global scope are accessible in the whole script and are called global variables. Local variables being restricted to a specific function block.

```
global_var = 123 # Free/Global variable

def my_function():
    local_var = "Hello" # Local variable
    print(local_var, global_var)

# The function can use both variables
my_function()

# This is not possible!
# print(local_var)
```

Hello 123

Keyword global

In some problem solutions it is meaningful to have a global variable which gets updated from within a local scope of a function.

```
counter = 10
print(counter)

def reset():
    global counter
    counter = 0
    print(f"Counter reseted: {counter}")

reset()
```

10

Counter reseted: 0

Note: It's best practice to avoid global variables.

Keyword `nonlocal`

The Python keywords `global` and `nonlocal` target different scope levels when modifying a variable from within a function.

```
def outer():
    enclosing_var = 20

    def inner():
        nonlocal enclosing_var
        enclosing_var = 200

        global global_var
        global_var = 100

        print(f"Inner: global_var={global_var}, enclosing_var={enclosing_var}")

    inner()
    print(f"Outer: global_var={global_var}, enclosing_var={enclosing_var}")

print(f"Start Global: {global_var}")
outer()
print(f"End Global: {global_var}")
```

```
Start Global: 123
Inner: global_var=100, enclosing_var=200
Outer: global_var=100, enclosing_var=200
End Global: 100
```

Parameter Value

Parameters are always references to arguments. An argument can be of a mutable (like lists) or immutable data type. The content of mutable data types can be “globally altered in functions.

```
def my_add(p):
    result = p
    result[-1] = 99 # Change last value to 99
    return result

var = [9,8,7,6,5,4,3,2,0] # Last value is 0
```

```
print(my_add(var))
print(var) # List was changed by the function
```

```
[9, 8, 7, 6, 5, 4, 3, 2, 99]
[9, 8, 7, 6, 5, 4, 3, 2, 99]
```

Unpacking Arguments (*args and **kwargs)

Python assigns one object to the first and the last variable and packs all the remaining objects into a tuple that is assigned to the variable with the packing operator

```
var1 = 1
var2 = ["hello", "-", "world"]
var3 = 22
print(var1, var2, var3)

# Using the packing operator
var1, *var2, var3 = 1, "hello", "-", "world", 22
print(var1, var2, var3)
```

```
1 ['hello', '-', 'world'] 22
1 ['hello', '-', 'world'] 22
```

- ***args**: packing operator (*) packs all non-keyword arguments into a tuple.
- ****kwargs**: double packing operator(**) packs all keyword arguments into a dictionary

```
def my_fun(*args, **kwargs):
    print(args); print(kwargs)

my_fun('hello', 'world', arg1=33, arg2=55)
```

```
('hello', 'world')
{'arg1': 33, 'arg2': 55}
```

In contrast, lists and tuples can be unpacked using the packing operator (*) and (**) for dictionaries, respectively.

Recursion

A recursive function is simply a function that calls itself during its execution. It is especially useful to loop through a data set with undefined dimension and undefined iteration count. Despite increased complexity and risks, recursion allows to solve complex problems in a short way.

```
def factorial(n):  
    if n == 1:  
        return 1  
    else:  
        return n * factorial(n-1)  
  
factorial(5)
```

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SW08: Object-Orientated Programming

Definition

Object-Oriented Programming (OOP) is a programming paradigm that structures code around objects rather than functions and logic. It's a powerful way to model real-world entities and relationships in your code, leading to more modular, reusable, and scalable applications.

Classes

Assembly instructions for box construction are represented by the class. The produced boxes, all with the same set of tools but different property values, are called objects. In Python everything is an object and variables hold references to these objects. This means, when you assign an object to a variable, you actually create a reference to that object in memory.

A class in python consist of several elements: - **header** with class name indicates the beginning of a class description. - **init()** method: The constructor constructs a new object of the class. - **variables**: class: Variables belonging to the class description. Object: variables belonging to a particular object. - **methods** (functions): Class: methods provided by the class description. Object: methods provided by the object. - **self** attribute: Reference to the object: required to access object attributes.

```
class MyClass(): # The class
    def __init__(self): # Initial function
        pass

    def my_function(): # Methode of the class object
        pass
```

Note: For class names, we use CamelCase instead of underscores, as in function names.

__init__

Generating a new object, requires calling the class name and defining the specific properties by attributes passed to the constructor. Each time when generating a new object, python calls the constructor and passes the “properties” (attributes) to the **init()** method.

Methodes and variables

The class definition fully describes the class (i.e. the object). It comprises a mix of all variables and methods provided for the whole class and particular objects. Methods and variables are called with the object or by reference (self-keyword), respectively. Similarly to variables, methods can belong to an object or a class itself. The declaration and call is the same as for regular functions, yet always requires a reference

```
class MyClass(): # The class
    def __init__(self): # Initial function
        pass

    def my_function(self): # Methode of the class object
        pass

    @classmethod
    def class_function(cls): # Methode only of of the class
        pass
```

Note: The keyword **cls** is a convention but not necessary.

self

The self keyword points to the object it belongs to (namely itself). It allows to call or apply a variable and method, respectively, of (on) a particular object.

Method types

Object Methods

Belong to a particular object. Modifying object's state.

```
def change_obj_state(self):  
    self.state = "New State"  
    return True
```

Class Methods

Belong to a particular class. Modifying class' state.

```
@classmethod  
def change_cls_state(cls):  
    cls.state = "New State"  
    return True
```

Static Methods

Don't belong to object nor class. Provide operations independent of any object or class state.

```
@staticmethod  
def utility_func():  
    print("Done")  
    return True
```

Dunder Methodes

Dunder Methods (or Magic methods), are special methods in Python classes marked by double leading and trailing underscores (e.g., `__init__`, `__str__`).

`__str__()`

By implementing a `str()` method, the programmer can define what is printed when an object of an own class is passed to the print function.

```
def __str__(self):  
    return self.cls_name
```

SW09: Object-Orientated Programming II

Inheritance

Inheritance creates a hierarchy of classes. This avoids code redundancy and increase maintainability. Use the function `isinstance(object, classinfo)` or `issubclass(class, classinfo)` to check if an instance is a subclass of a other class. Child classes get all properties (attributes) from parent class. This means, class or object variables or methods defined in the parent class also hold in the child classes. Typically, child classes extend parent classes. In some cases, a child class can override variables or methods from parent class.

```
class Animal:  
    def __init__(self, weight):  
        self.weight = weight  
  
class Bird(Animal):  
    def __init__(self, weight):  
        super().__init__(weight) # Inheritance from the parent class
```

When extending functionality of parent class by overriding a method (or in general to use parent's methods), they can be referred to using `super()` or parent's class name.

```
class Animal:  
    def __init__(self, weight):  
        self.weight = weight  
  
    def get_weight(self):  
        return self.weight  
  
class Bird(Animal):  
    def __init__(self, weight):  
        super().__init__(weight)  
  
    def get_weight(self):  
        return super().get_weight()/10
```

The Class “object”

This is the ultimate base class of all other classes. It has methods that are common to all instances of Python classes. When the constructor is called, it returns a new featureless object. The constructor does not accept any arguments.

Multiple Inheritance

- Single inheritance: **one** child class inherits from **one** parent class.
- Multiple single inheritance: **multiple** child classes inherit from **one** parent (the same) class.
- Multiple inheritance: **one** child class inherits from **multiple** parent classes.

Method Resolution Order (MRO)

The Method Resolution Order (MRO) is the order in which Python searches for methods and attributes in a class hierarchy. It dictates the sequence of base classes that are checked when a method is called on an object of a derived class, especially in cases of multiple inheritance (where a class inherits from more than one parent). The MRO is essential for making multiple inheritance reliable and predictable.

Note

The `super()` function relies entirely on the MRO. When you call `super().method()`, Python uses the MRO list to find the next class in the hierarchy after the current class that implements `method()`, ensuring that base class methods are called in the correct, standardized order.

Access Modifiers

Since Python is a script language and does not compile code in advance to execution, it lacks of keywords defining attribute’s accessibility. Yet, there is a convention in the naming of attributes using `_` for protected and for `__` private attributes for treating access restrictions.

Note: This is only a convention. Python has no hard restrictions and allows to access all attributes on object level.

SW10: Modules and Packages

Modules

A module is a file containing Python definitions and statements. We can use modules for better structuring the source code of the whole application.

Import

In order to access content of other modules, they can be imported to a module as whole or parts of it.

```
import numpy           # Declares reference to module
import numpy as np      # Renames reference to
from numpy import array # Declares reference to specific attribute
from numpy import *     # Declares references to "all" attributes
```

When imported, attributes of external modules are accessed differently, depending on the import.

Note

When it comes to aliases for frequently used modules, there are conventions.

```
import numpy as np
import pandas as pd
from matplotlib import pyplot as plt
```

Referring to external modules and files require either absolute or relative references. By default, root is set to the main executed script. The main executed script only allows absolute path. By default, imported modules are searched relative to the script path. For this case, the module `sys` allows to extend the search space for imported modules.

```
sys.path.append('C:\\Users\\additional\\location') # Windows
sys.path.append('/home/username/additional/location') # Linux
```

If `__name__ == "__main__":`

The If `__name__ == "__main__":` block determines how a Python file is being executed.

- If the file is run directly (as the main script), the variable `__name__` is set to `"__main__"`, and the code inside the block runs. This is where you put your application's main execution logic (e.g., calling your primary function).
- If the file is imported as a module into another script, `__name__` is set to the module's name, and the code inside the block is skipped. This allows the file to be used as a reusable library without triggering unwanted side effects.

This structure allows a single Python file to be both runnable and importable.

Packaging

Python enables to structure project resources using packages. a folder becomes a package as soon as it comprises a script named `__init__.py`

Note: Since Python 3.3, there has been no need to declare an `__init__.py` file for packages, but it is still good practice.

The `__init__.py` Script

Objects defined in the `init.py` are bound to names in the package's namespace.

External Packages

The Python Software Foundation hosts an own repository for official Python packages called [PyPI](#). `pip` is a command-line interface (CLI) tool for installing Python packages.

```
pip install <packages>
pip uninstall <packages>
pip list
pip freeze # Output in requirement format
```

SW11: Must have Data Science Packages

Data Manipulation

Numpy

NumPy is the fundamental package for scientific computing in Python. It is a Python library that provides a multidimensional array object, various derived objects, and an assortment of routines for fast operations on arrays.

Multidimensional Array objects

Multidimensional array object (**ndarray**) are sequences, matrices or multidimensional matrices of numerical values. Important differences between NumPy arrays and the standard Python sequences are:

- Fixed size at creation. Changing size of **ndarray** creates new array.
- All elements are required to be of the same data type.
- NumPy arrays facilitate advanced mathematical and other types of operations on large numbers of data.

Pandas

Pandas is a fast, powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language.

- **Fast DataFrame Object:** A core, efficient object for data manipulation with **integrated indexing**.
- **I/O Capabilities:** Easy **reading and writing** of data across various formats (CSV, Excel, SQL, HDF5).
- **Data Handling:** **Intelligent data alignment** and integrated **missing data handling**.
- **Aggregation:** A powerful **group by engine** for split-apply-combine operations.

Data Frames

Data frames are tabular data structure commonly used in data analysis and statistical computing.

- Columns represent variables (features).
- Rows represent observations (records).
- Indexing by labels or numerical indices: [row, col]

Note

A series is a one dimensional data (index and one column of a data frame).

Data Visualization

Matplotlib

Matplotlib is a low-level library that provides a lot of flexibility and control over the creation of plots. Matplotlib has three fundamental components used for creating and managing plots:

- Pyplot: a collection of functions to create figures.
- Figure object: represents the entire figure in which you can plot one or more axes.
- Axes object: represents a single or a set of plots within a figure. This is the area where data is plotted.

Seaborn

Built on top of Matplotlib, Seaborn is a high-level library specifically designed for statistical data visualization. Since seaborn builds on top of matplotlib, it is mandatory to import pyplot from matplotlib in order to add titles or show the plot directly.

Jupyter Notebook

Jupyter Notebook is an open-source web application that allows you to create and share documents that contain code, equations, visualizations, and narrative text. It is widely used for data analysis, scientific research, machine learning, and educational purposes.

- Interactive Code Execution: write and execute code in a cell-based format, allowing for iterative development and testing.
- Rich Text Support: include [Markdown](#) for formatting text, including headings, lists, links, and images.
- Visualizations: create and display plots and figures inline, making it easy to visualize data alongside your code.
- Report: export the notebook as HTML or PDF, allows for having a report always ready.

SW12: Short Functions

Deep vs. Shallow Copy

Python only has objects and references to them. So, variables are always references to objects stored in the memory in one or more storage cells. Consequently, when copying a variable, we copy the reference to a particular object – not the object itself.

```
# Define a var
x = 123
y = x # Copy x to y

# Both vars have the same memory location
print(id(x))
print(id(y))
```

```
11759592
11759592
```

To create a copy of a sequence object with a new reference (ie. `id`), sequence objects implement the `copy()` method.

Note: The method `copy()` is only available for mutable objects.

Copying a sequence returns a shallow copy of the object only. References to sub-sequences still remain the same and are affected by the same side-effect as when duplicating the reference by reassignment.

Lambda Function

Concept

When associating a function to a variable, this can be done directly without function declaration first. Associated to a variable, they can be called as regular functions by the variable as reference instead of the function name. are typically used for expressions in combination with filter functions such as for the `map()` or `sort()` functions.

`map()`-Function

Applies a function to every single item of an iterable.

```
my_list = [2, 4, 6, 8, 10]
# Divide every value from the list by 2
new_list = list(map(lambda x: int(x/2), my_list))
print(my_list)
print(new_list)
```

```
[2, 4, 6, 8, 10]
[1, 2, 3, 4, 5]
```

sort()-Function

Sorts an iterable based on a given comparator.

```
my_list = [4, 10, 6, 2, 8]
# Sort the values in reverse order
my_list.sort(key=lambda x: -x)
print(my_list)
```

[10, 8, 6, 4, 2]

List Comprehension

Instead iterating a list with a for loop for applying a single expression on each element, a single expression can be applied in a simpler way on one line using a list comprehension.

```
my_list = [1, 2, 3]
new_list = []
# Iterate over a list
for value in my_list:
    new_list.append(value * 2)
print(f"Original list: {my_list}")
print(f"New list with loop: {new_list}")

### IS THE SAME AS ###

my_list = [value * 2 for value in my_list]
print(f"New list with comprehension {my_list}")
```

Original list: [1, 2, 3]

New list with loop: [2, 4, 6]

New list with comprehension [2, 4, 6]

Note

Comprehensions can be applied in-place

Comprehensions only apply a single expression on each list element. However, it provides the option defining conditions for the elements on that it should be applied.

```
my_list = [1, 2, 3, 4]
# Only double even values from the list
my_list = [value * 2 if value % 2 == 0 else value for value in my_list]
print(my_list)
```

[1, 4, 3, 8]

Note: The single expression can also be a regular function

Nested Loops

comprehension allow to apply expressions to multi-dimensional lists – nested list comprehension.

```
my_list = [[1, 1], [2, 2], [3, 3]]
new_list = []
# Iterate over both lists
for coordinate in my_list:
    new_coordinate = []
    for entry in coordinate:
        new_coordinate.append(entry * 2)
    new_list.append(new_coordinate)
print(f"Original list: {my_list}")
print(f"New list with loop: {new_list}")

### IS THE SAME AS ###

my_list = [[entry * 2 for entry in coordinate] for coordinate in my_list]
print(f"New list with comprehension {my_list}")
```

Original list: [[1, 1], [2, 2], [3, 3]]
 New list with loop: [[2, 2], [4, 4], [6, 6]]
 New list with comprehension [[2, 2], [4, 4], [6, 6]]

Tuples

Despite the similarity between tuple and list in Python, there exist no tuple comprehension. Generating a tuple with a comprehension requires an explicit typecast `tuple()` or a tweak using unpacking operator for variable assignment.

```

my_tuple = (1, 2, 3, 4)

x = tuple([i*2 for i in my_tuple])
print(f"With typecast: {x}")

y = *[i*2 for i in my_tuple],
print(f"With unpacking operator: {y}")

```

With typecast: (2, 4, 6, 8)
 With unpacking operator: (2, 4, 6, 8)

Dictionary

Dictionary comprehension requires the format of dictionary elements as expression results.

```

x = [1, 2, 3, 4, 5]

d1 = {i:i for i in x}
print(d1)
d2 = {k:v for (k,v) in enumerate(x)}
print(d2)
d3 = {k:v for (k,v) in zip(["a", "b", "c", "d", "e"], x)}
print(d3)

```

```

{1: 1, 2: 2, 3: 3, 4: 4, 5: 5}
{0: 1, 1: 2, 2: 3, 3: 4, 4: 5}
{'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}

```

Type Annotation

The Python runtime does not enforce function and variable type annotations. They can be used by third party tools such as type checkers, IDEs, linters, etc. In particular, we only define the type (any Python class) of references separated by colon (:) and arrows (->).

```

x: int = 10 # Var x is an integer
def my_function(x: int) # The param x has to be an integer
def my_function(x) -> int: # The function returns an integer

```

Assert Function

Python's `assert` statement allows you to write sanity checks in your code. These checks are known as assertions, and you can use them to test if certain assumptions remain true while you're developing your code. If any of your assertions turn false, it indicates a bug by raising an `AssertionError`.

```
def division(a, b):
    assert b != 0, "Division by Zero" # Divider can't be Zero

    res = a / b
    return res
```

SW 14: Exceptions, Generators, Decorators

Exceptions

```
try:
    num = input('Enter a number: ')
    frac = 1/int(num)
except ValueError as e:
    print('no number given:', e)
except ZeroDivisionError:
    print('can not divide by 0')
except:
    print('any other error happend')
else:
    print(f'the fraction is: {frac}')
finally:
    print('leaving try-except clause')
```

Custom Error

```
class CustError(Exception):
    def __init__(self, message):
        super().__init__(message)
        # customize error code...
        self.errors = "custom error"
```

An error can be raised at any position in the script using the keyword: `raise`

```
if input('enter number: ') == '0':  
    raise CustError('customized error message')
```

Assert Statement

`assert` can be useful for sanity checks in development, testing, and debugging phase but can get optimized away in production code (proper exception handling).

```
x = 21  
assert isinstance(x, int), 'Value should be of type int'
```

Generators

Imagine you have a list with a billion numbers. If you save this as a normal list, your working memory (RAM) would immediately become full. An iterator, on the other hand, only calculates or loads the element that is currently needed. The generator function is like a regular function, yet using the `yield` keyword instead of `return`. To understand the concept in Python, this distinction helps:

- Iterable: An object that you can use in a for loop (e.g., list, string, dictionary). It has the `iter` method.
- Iterator: The object that actually does the work. It has the `next` method and returns the values to you one after the other.

Decorators

Technically speaking, a decorator is a function that takes another function as an argument, extends its behavior, and then returns it.

```
import time  
from functools import wraps  
  
def timer(func):  
    """A decorator that measures the execution time of a function."""  
    @wraps(func) # Best practice: keeps the original function's name and docstring  
    def wrapper(*args, **kwargs):  
        # 1. Action before the function call  
        start_time = time.time()
```

```

    # 2. Execute the actual function
    result = func(*args, **kwargs)

    # 3. Action after the function call
    end_time = time.time()
    duration = end_time - start_time
    print(f"Duration of {func.__name__}: {duration:.4f} seconds")

    # 4. Return the result of the original function
    return result
return wrapper

@timer
def long_computation():
    """Simulates a heavy task."""
    time.sleep(2)
    print("Computation finished!")

# Calling the decorated function
long_computation()

```